

HIGHER TIER ONLY QUESTIONS

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
2	(a)			$\text{CO}_2 + \text{C} \rightarrow 2\text{CO}$ award (1) for both reactants award (1) for product and balancing	2			2		
	(b)	(i)		2 $\text{Fe}_2\text{O}_3 + 3\text{C} \longrightarrow$ 4 $\text{Fe} +$ 3 CO_2		1		1		
		(ii)		<u>iron oxide</u> is reduced $\Rightarrow$ it loses oxygen (1) neutral answer – <u>iron</u> is reduced $\Rightarrow$ it loses oxygen carbon is oxidised $\Rightarrow$ it gains oxygen (1) if no other credit award (1) for generic description or for identifying which reactants are oxidised and reduced - oxidation is gaining oxygen, reduction is losing oxygen - iron oxide is reduced (to form iron) and carbon is oxidised (to form carbon dioxide) - oxygen goes from iron oxide to carbon reference to electrons is neutral		2		2		
	(c)	(i)		decomposition	1			1		1
		(ii)		CaSiO_3	1			1		
				Question 2 total	4	3	0	7	0	1